

The Invisible Spotlight:Reconstructing Fan Votes and Optimizing Scoring Mechanisms

Summary

The binary paradox between technical fairness and commercial value lies at the core of the dilemma facing reality television programs such as Dancing with the Stars (DWTS). The evolutionary trajectory of the competition format over 34 seasons underscores the inherent challenges in integrating professional evaluation with public engagement. This paper presents a systematic framework designed to reconstruct implicit voting data, diagnose mechanical biases, deconstruct preference logic, and propose an optimal solution.

To reconstruct the "black box" of fan voting, we propose the **Inverse Bayesian Reconstruction Model (IBRM)**. Addressing the ill-posed inverse problem where only elimination outcomes are observable, we employ **Monte Carlo Rejection Sampling** to approximate the posterior distribution of fan votes. This model transforms "inverse uncertainty" into statistical variance and defines a "Survivor Ambiguity" mechanism, successfully reconstructing the latent voting data.

To diagnose the root causes of inequity, we construct a **Counterfactual Simulation Engine** and select five representative contestants for simulation. Through historical reenactments under varying rules, we identify the "Variance Imbalance Principle" and quantify it using an Elite Index, demonstrating that the ranking system possesses more robust constraint mechanisms, while the "Save" mechanism offers a protective buffering effect.

To analyze the impact of various features on competition outcomes, we employ **Ant Colony Optimization (ACO)** for objective feature selection and utilize **Linear Mixed Models (LMM)** to decompose hierarchical effects, developing a hybrid ACO-LMM framework. The results reveal a critical divergence: judges prioritize the logic of competitive sports, whereas fans favor "underdog" narratives. This axiological misalignment is identified as the sociological root of format failure.

Finally, we design the **Grand Prix Dynamic (GPD)** System. This model introduces an F1-style non-linear point mapping and utilizes Sigmoid dynamic weighting to achieve a smooth transition from an entertainment orientation in the early stages to a competitive orientation in the later stages. Additionally, a "Dance-Off" mechanism is incorporated to enhance viewership through high-pressure, immediate confrontation.

Validation through ablation studies indicates that the GPD System achieves global Pareto optimality. It maintains high Fan Efficacy (**0.76**) while ensuring Professional Elitism (**0.95**). The "Bobby Bones Stress Test" further confirms the system's robustness in balancing commercial success with competitive integrity.

Keywords: IBRM; Counterfactual Simulation ; ACO ; LMM ; Pareto Optimality

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1 Introduction

1.1 Problem Background

Dancing with the Stars, originating from Britain's Strictly Come Dancing, has aired 34 seasons in its U.S. version. The core of the show is celebrities pairing with professional dancers for competition, with results determined by a dual mechanism—expert judges' scores and fan votes—balancing technical skills and popularity.

To balance professional impartiality and public participation, the production team has adjusted the scoring mechanism three times: Seasons 1-2 adopted a ranking-based points system, where contestant Jerry Rice advanced with high fan votes despite low judges' scores, sparking controversy; Seasons 3-27 switched to a percentage-based system to increase judges' weight, yet contestant Bobby Bones still claimed the championship through fan votes; since Season 28, the ranking-based system has been restored, alongside the new "Judges' Save" segment (judges decide the elimination of the bottom two contestants).

This evolution of the competition system highlights the core contradiction of reality shows, providing an important context for using mathematical modeling to deconstruct voting mechanisms, reconstruct latent variables, and optimize the competition system.

1.2 Restatement of the Problem

Given the background information of this problem, our core tasks are as follows:

- **Estimation of Latent Fan Votes:** Establish a mathematical model to inversely infer the weekly unpublished fan votes of each contestant, verify the accuracy of the model in explaining elimination results, and quantify the statistical certainty of the estimation outcomes;
- **Comparison of Competition Systems and Counterfactual Simulation:** Compare the differences between the two aggregation methods—"ranking-based system" and "percentage-based system"—and analyze their bias towards fan voting weight. For controversial historical cases, simulate the potential changes in final rankings and elimination results under different scoring rules and the "Judges' Save" mechanism;
- **Analysis of Factors Influencing Competition Outcomes:** Using the estimated data, quantify the impact of celebrity characteristics (such as age and occupation) and the identity of professional dance partners on competition performance, and reveal the distinct mechanisms of action of these factors in the two dimensions of judges' scores and fan votes;
- **Design of a New Scoring System:** Propose a new scoring and elimination mechanism that balances fairness and program viewership, demonstrate the advantages of the new scheme compared to the existing rules, and provide specific implementation recommendations to the program producers;

1.3 Our Work

The work undertaken in relation to this problem is primarily illustrated in Figure 1.

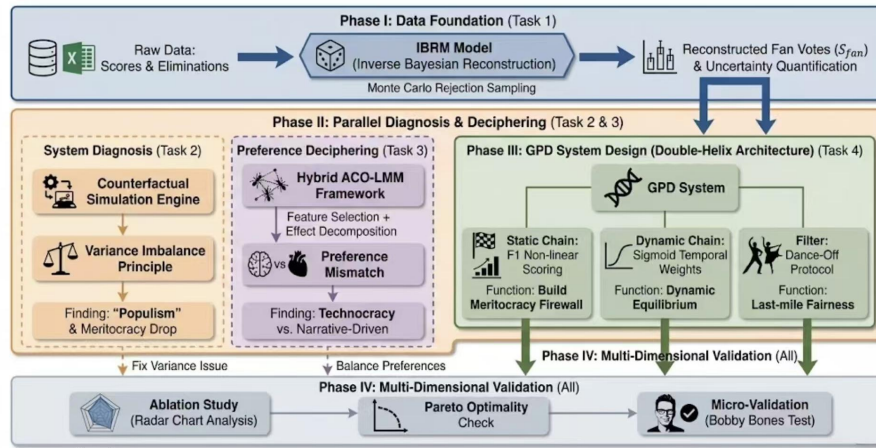


Figure 1: Our Work

2 Data processing

2.1 Assumptions and Justifications

In this study, we propose the following core assumptions:

- **Assumption 1:** Determinism of Elimination and Rule Consistency

To facilitate the construction of the Inverse Bayesian Reconstruction Model (IBRM), we assume that the weekly elimination results strictly adhere to the calculation rules published by the show, with no undisclosed human manipulation or random intervention.

- **Assumption 2:** Relative Comparability Across Seasons

To establish a unified mathematical foundation for data analysis, we assume that through the Universal Percentile Harmonization (UPH) transformation, scores from different systems can be mapped into a unified space for comparison.

2.2 Data Integration & Feature Engineering

For the original dataset, we first excluded samples of non-natural withdrawals to avoid bias from uncorrelated factors. For missing judge sub-scores, mean imputation was adopted, as it is suitable for continuous data with random missing patterns and can retain the overall distribution characteristics of judge scores. Additionally, Nielsen historical ratings were introduced as external variables, and aligned with the core dataset using *season-week-contestant ID* as the key to verify the correlation between contestant popularity (reflected by votes) and commercial value.

Furthermore, natural language processing (NLP) techniques were utilized to cluster contestant occupations into 5 meta-categories (e.g., Arts, Sports, Entertainment, Business, and Others). Specifically, we extracted occupation keywords, calculated semantic similarity via TF-IDF and cosine distance, and performed K-means clustering ($K = 5$, determined by elbow method). This clustering effectively reduces feature sparsity and improves the convergence efficiency of the Linear Mixed Model (LMM) used in subsequent analyses.

2.3 Universal Percentile Harmonization

To resolve the dimensional incompatibility between the Rank System and Percentage System adopted in DWTS history, we proposed the UPH framework, which maps all dependent variables into a unified 0-100 percentile space. For contestant i in week t of season s , the standardized score $S_{i,t,s}^*$ is defined as:

$$S_{i,t,s}^* = 100 \times \frac{N_{t,s} - r_{i,t,s}}{N_{t,s} - 1} \quad (1)$$

where $N_{t,s}$ denotes the total number of contestants in week t of season s , and $r_{i,t,s}$ represents the relative rank of contestant i (rank=1 for the highest score, rank= $N_{t,s}$ for the lowest). This transformation has two core properties:

- **Scale Invariance:** It eliminates differences in scoring rules, ensuring that $S_{i,t,s}^*$ consistently represents the relative competitive level of contestants, regardless of whether the original system is Rank or Percentage.
- **Dynamic Adaptation:** It automatically corrects the probability space bias caused by the progression of the competition (i.e., $N_{t,s}$ decreases as contestants are eliminated each week). This adaptability ensures that scores across different weeks and seasons are comparable.

The UPH framework establishes a unified mathematical foundation for the longitudinal comparison of data across 34 seasons, which is crucial for the validity of subsequent cross-season analyses.

3 Inverse Vote Reconstruction

3.1 A Bayesian Perspective

The primary task of this study is to address the data missing problem, namely, inferring the underlying fan voting distribution S_{fan} from the known observed data (judge scores S_{judge} and elimination results E). Mathematically, this is a classic *ill-posed inverse problem*.

$$\text{argmin} (w \cdot S_{\text{judge}} + (1 - w) \cdot S_{\text{fan}}) = E_{\text{observed}} \quad (2)$$

Here, w denotes the weight. Since $\text{argmin}(\cdot)$ is a nonlinear and irreversible operator, there exist infinitely many combinations of S_{fan} that can lead to the same elimination

result. For example, the elimination of a high-scoring contestant may stem from either an extremely low fan vote share (e.g., 3%) or a moderately low share (e.g., 5%) coupled with outstanding performances from opponents, and analytical solutions cannot distinguish such subtle differences.

To quantify this inherent uncertainty, this study constructs a *Bayesian statistical inference* framework. Our goal is not to seek a single deterministic solution, but to derive the *posterior distribution* of the fan voting vector.

According to Bayes' theorem, the posterior distribution $P(\mathbf{V}_F | \mathbf{S}_J, e^*)$ of the parameter $\mathbf{V}_F$ can be expressed as:

$$P(\mathbf{V}_F | \mathbf{S}_J, e^*) = \frac{P(e^* | \mathbf{V}_F, \mathbf{S}_J) \cdot P(\mathbf{V}_F)}{P(e^* | \mathbf{S}_J)} \quad (3)$$

The physical meaning and mathematical form of each term are as follows:

Prior Distribution $P(\mathbf{V}_F)$: This term represents our prior belief about the fan voting distribution before observing the elimination result.

$$\mathbf{V}_F \sim \text{Dirichlet}(\alpha_1, \dots, \alpha_N) \quad (4)$$

where the parameter vector α controls the sparsity of the distribution. Setting $\alpha < 1$ can effectively simulate the concentration trend of votes.

Likelihood Function $P(e^* | \mathbf{V}_F, \mathbf{S}_J)$:

The likelihood function describes the consistency between the model and reality:

$$P(e^* | \mathbf{V}_F, \mathbf{S}_J) = \mathbb{I} \left(\underset{i}{\operatorname{argmin}} (w \cdot \text{Rank}(\mathbf{S}_{J,i}) + (1 - w) \cdot \text{Rank}(\mathbf{V}_{F,i})) = e^* \right) \quad (5)$$

That is: If the simulated elimination result caused by the parameter combination $\mathbf{V}_F$ is consistent with the historical record e^* , the likelihood is 1; otherwise, it is 0.

Marginal Likelihood $P(e^* | \mathbf{S}_J)$:

In summary, the posterior distribution of fan voting is proportional to the product of the prior and the indicator function:

$$\underbrace{P(\mathbf{V}_F | D_{obs})}_{\text{Posterior}} \propto \underbrace{\mathbb{I}(\text{GameLogic}(\mathbf{V}_F, \mathbf{S}_J) = e^*)}_{\text{Likelihood Constraint}} \cdot \underbrace{P(\mathbf{V}_F)}_{\text{Prior}} \quad (6)$$

3.2 Monte Carlo Rejection Sampling

Due to the posterior distribution containing the nonlinear indicator function $I(\cdot)$ and high-dimensional simplex constraints, analytical solutions are extremely difficult. Therefore, this study adopts the Monte Carlo Rejection Sampling algorithm for numerical approximation.

In this model, we set the prior distribution $P(\mathbf{V}_F)$ as the proposal distribution, the posterior distribution $P(\mathbf{V}_F | D_{\text{obs}})$ as the target distribution, and the envelope constant to $M = 1$.

For a sample $\mathbf{v}^{(i)}$ drawn from the prior distribution, we accept the sample with probability A :

$$A = \frac{\tilde{p}(\mathbf{v}^{(i)})}{Mq(\mathbf{v}^{(i)})} = \frac{\mathbb{I}(\text{Consistent}) \cdot P(\mathbf{v}^{(i)})}{1 \cdot P(\mathbf{v}^{(i)})} = \mathbb{I}(\text{Consistent}) \quad (7)$$

That is, the sample is accepted if and only if it is consistent with the historical result.

The detailed computational process of this model is presented as *Algorithm1* in pseudocode:

Algorithm 1 Monte Carlo Rejection Sampling for Fan Vote Reconstruction

Require: Judge Scores matrix $\mathbf{S}_J$; Observed Elimination e^* ; Season Rules $\mathcal{R}$; Iteration Count K ; Prior Parameter α

Ensure: Set of Valid Fan Vote Samples Ω_{valid}

```

1:  $\Omega_{\text{valid}} \leftarrow \emptyset$  ▷ Initialize empty set
2: for  $k = 1$  to  $K$  do
3:   Step 1: Proposal Generation
4:    $V_{\text{sim}} \sim \text{Dirichlet}(\alpha)$  ▷ Sample from prior
5:   Step 2: Simulation
6:    $R_{\text{comb}} \leftarrow \text{ComputeRank}(\mathbf{S}_J, V_{\text{sim}}, \mathcal{R})$ 
7:    $e_{\text{sim}} \leftarrow \text{GetEliminated}(R_{\text{comb}})$  ▷ Identify lowest scorer
8:   Step 3: Rejection Criterion
9:   if  $e_{\text{sim}} == e^*$  then ▷ Check consistency with history
10:     $\Omega_{\text{valid}} \leftarrow \Omega_{\text{valid}} \cup \{V_{\text{sim}}\}$  ▷ Accept sample
11:   end if
12: end for
13: return  $\Omega_{\text{valid}}$ 

```

According to the mathematical properties of rejection sampling, when the number of iterations K is sufficiently large, the empirical distribution of accepted samples will converge to the target posterior distribution. The mathematical proof is as follows:

$$P(\theta_{\text{accepted}}) = \frac{q(\theta) \cdot A}{\int q(x) A dx} = \frac{P(\theta) \cdot \mathbb{I}(\text{Consistent})}{Z} = P(\theta | D_{\text{obs}}) \quad (8)$$

This proves that the sample mean μ and variance σ^2 obtained via rejection sampling are **Unbiased Bayesian Estimations** of the true historical voting outcomes. Using this method, we transform the uncertainty of the "inverse problem" into statistical **Variance**.

3.3 Result Analysis: Decoding Hidden Votes

After establishing the inverse vote reconstruction model, this section will delve into the reliability of estimation results, quantify the inherent uncertainty structure, and conduct

multi-dimensional validation of the model through historical consistency tests and case micro-analyses.

3.3.1 Historical Consistency Validation

To quantitatively evaluate the validity of model estimations, we define the **Posterior Predictive Consistency Ratio** (C_{rate}) as the core metric:

$$C_{\text{rate}} = \frac{1}{T} \sum_{t=1}^T \mathbb{I} \left(e_t^* \in \text{Bottom}_k \left(\hat{S}_{\text{total},t} \right) \right) \quad (9)$$

where $T = 264$ denotes the total number of elimination weeks. This metric quantifies the probability that the actual eliminated contestant falls into the model-predicted "danger zone" (the bottom k contestants).

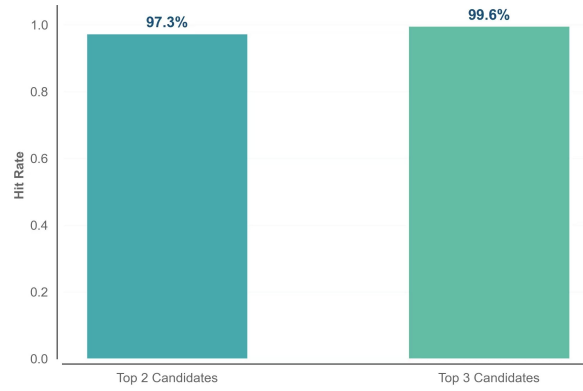


Figure 2: Global Posterior Accuracy

As illustrated in Figure 2, the global consistency ratio of the model reaches as high as 97.35% (when $k = 3$). This means that among the 264 elimination weeks, the IBRM model successfully identified the actual eliminated contestant as one of the three most dangerous contestants in 257 weeks. The remaining 2.65% of unrepresented weeks represent statistical **Anomalies**—namely, "Shock Eliminations". In these cases, the eliminated contestant was statistically in a safe zone but was still eliminated, indicating that the fan voting distribution in that week significantly deviated from the standard long-tailed prior assumption. The extremely high C_{rate} confirms that the likelihood function effectively filtered out unrealistic voting scenarios, ensuring the posterior distribution is highly consistent with historical reality.

3.3.2 Uncertainty Quantification

The uncertainty of estimations exhibits significant **Heteroscedasticity**, meaning that uncertainty is not uniformly distributed across contestants but systematically depends on their safety margins. We adopt the posterior standard deviation σ as the metric for measuring uncertainty.

As illustrated in Figure 3, the posterior standard deviation distribution across 34 seasons exhibits a significant right-skewed characteristic. The peak on the left side of the distribution corresponds to eliminated contestants or those on the verge of elimination. For such contestants, the elimination rule imposes a strong constraint—their total score must be strictly below a certain critical threshold. This hard boundary greatly compresses the feasible solution space, locking the estimated values of fan votes within an extremely narrow confidence interval. In contrast, the long tail on the right side of the distribution corresponds to contestants who advanced safely, with their uncertainty increasing significantly.

To investigate the driving mechanism of this distribution, we examined the joint distribution of estimated popularity (mean μ) and uncertainty (standard deviation σ) (Figure 4), which reveals a significant positive correlation between the two. This study defines this phenomenon as the **"Survivor Ambiguity"** mechanism, whose mathematical origin lies in the **Asymmetry of Constraints**.

Specifically, elimination acts as a hard constraint, providing explicit "upper bound information" on scores and thus effectively eliminating uncertainty. In contrast, survival serves as a soft constraint with high redundancy. For a contestant with high judge scores, whether they receive 20% or 50% of the fan votes, their "advancement" outcome remains

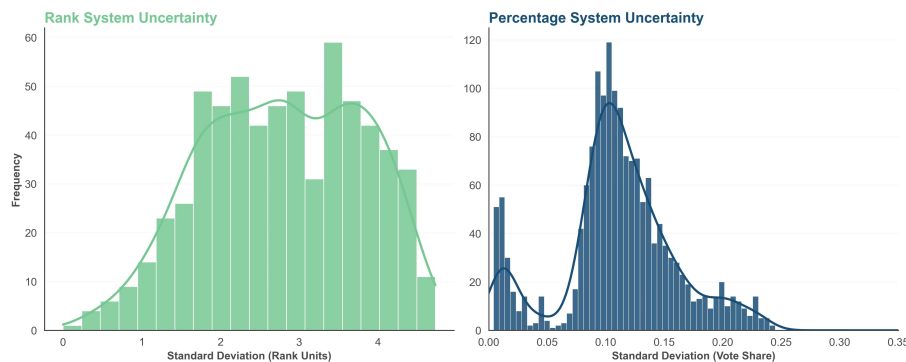


Figure 3: Global Distribution of Estimation Uncertainty

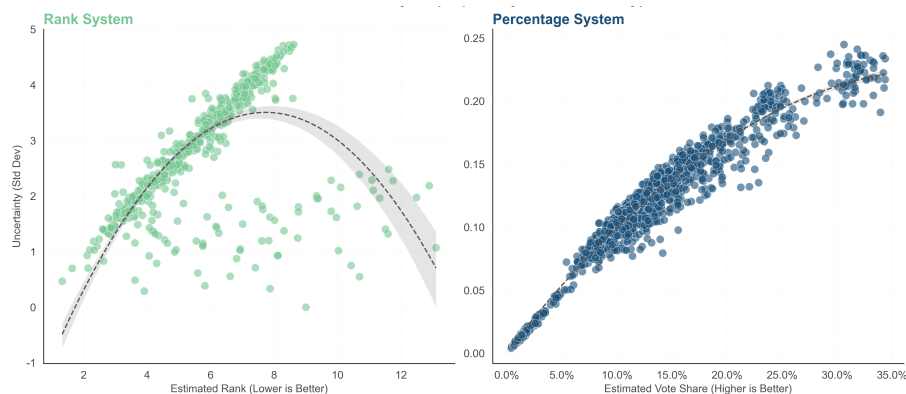


Figure 4: Correlation between Popularity and Uncertainty

unchanged. Since historical data only records the "advancement" status and cannot mathematically distinguish between "barely safe" and "absolutely safe," the model must retain a broader set of posterior samples in Bayesian inference, resulting in higher estimation variance.

In summary, this variation in uncertainty is not a flaw of the model but a faithful reflection of the inherent **Information Entropy** in the system. The model can precisely lock in the vote interval for losers but can only provide range estimates for winners. This demonstrates that the IBRM model does not fabricate precision in regions unsupported by data.

3.3.3 Case Study: Season 27, Week 4

To demonstrate the micro-level explanatory power of the IBRM model, we conduct an in-depth analysis of one of the most controversial moments in DWTS history—the elimination of Tinashe in Week 4 of Season 27.

Background Analysis: Week 4 presented a stark contrast between two typical contestants. Tinashe ranked 1st with a judge score of 26/30, while Bobby Bones hovered near the bottom of the leaderboard (rank 8) with a score of 20/30. Intuitively, Tinashe appeared to be in an absolutely safe zone. However, historical reality shows that she was eliminated, while Bobby Bones advanced safely.

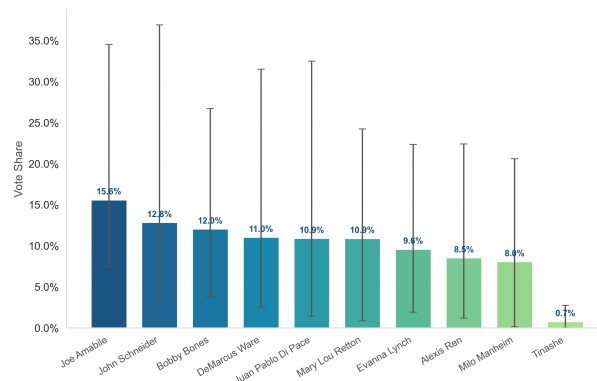


Figure 5: Reconstructed Vote Share for Season 27, Week 4 (with 95% Confidence Intervals)

Figure 5 presents the reconstructed fan voting distribution for Week 4 of Season 27, revealing a massive hidden deficit that judge scores failed to capture:

Tinashe's estimated fan vote share $\approx 0.7\%$ (95% CI: [0.023%, 2.723%]), while Bobby Bones' estimated fan vote share $\approx 12.0\%$ (95% CI: [3.76%, 26.703%]). Their confidence intervals do not overlap, indicating that Bobby Bones received approximately 17 times as many votes as Tinashe.

Thus, even though Tinashe was the technical superior, her marginal judge advantage was mathematically insufficient to offset even a small portion of Bobby's fan support surplus. The model's estimation of Tinashe's extremely low vote share was not a random guess but the only mathematical solution that satisfied the constraint of her elimination.

This not only reconstructs the historical truth but also deeply reveals the institutional reason why meritocracy was overwhelmed by surging popularity under this competition system.

4 Anatomy of Controversy: A Counterfactual Diagnostic Framework

4.1 Methodology: The Counterfactual Simulation Engine

To systematically address the core question of whether inherent biases exist in the existing competition system, this study constructs a **Counterfactual Simulation Engine**. Built upon the judge score matrix (S_{judge}) and fan vote matrix (S_{fan}) reconstructed in Task 1, this engine generates competition outcomes in a "parallel universe" by permuting the calculation rules (processing logic) while keeping input data unchanged. Specifically, we designed four experimental scenarios (as shown in Table 4-1), corresponding to different historical phases and hypothetical reform proposals. By comprehensively backtesting historical data from 34 seasons, this study aims to isolate the **Net Causal Effect** of rule changes on contestant rankings.

Table 1: Experimental Scenarios and Corresponding Rules

Experimental Scenario	Rule
Model A: Rank Only (S1–S2)	Total score is determined by the sum of ranks
Model B: Percentage Only (S3–S27)	Total score derives from weighted judge/fan score percentages
Model C: Rank + Save (S28+)	Introduces the judges' save mechanism
Model D: Percentage + Save	Hypothetical hybrid rule

To quantify the effectiveness of each system in maintaining competitive fairness, this study defines the **Meritocracy Index (M)**. This metric calculates the **Spearman's Rank Correlation Coefficient** between the simulated final ranking ($R_{\text{final},i}$) and the judge-average ranking ($R_{\text{judge},i}$): $M = 1 - \frac{6 \sum d_i^2}{n(n^2-1)}$

where $d_i = R_{\text{final},i} - R_{\text{judge},i}$ is the rank difference, and n is the total number of contestants. An M value approaching 1 indicates that the system maximizes the principle of "the most technically skilled contestants win," whereas a lower M implies that entertainment factors dominate competition outcomes. Figure 7 presents the fairness scores of the four systems, showing that the "Rank + Save" combination achieves the highest score ($M \approx 0.62$), significantly outperforming the "Percentage Only" mechanism ($M \approx 0.405$). This finding initially identifies the latter as the institutional root cause of frequent historical controversies.

4.2 The Variance Game between Percentage and Rank Systems

From Figure 6, we can observe that the percentage system exhibits a significant and systematic bias toward fan votes. By comparing the simulation results of the rank-based

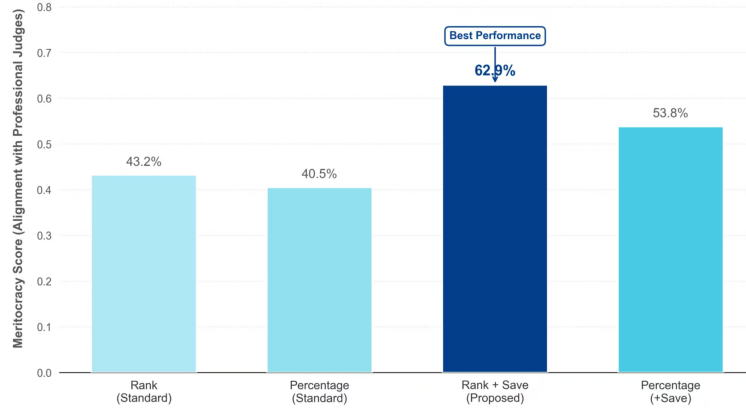


Figure 6: Comparison of Meritocracy Indices Across Four Scoring Systems

method and the percentage-based method, the data reveals the underlying mathematical mechanism causing competition system bias: the Variance Imbalance Principle. Among the 17 cases where the two methods diverged, the percentage system eliminated contestants with fewer audience votes, while the rank system eliminated those with lower judge scores.

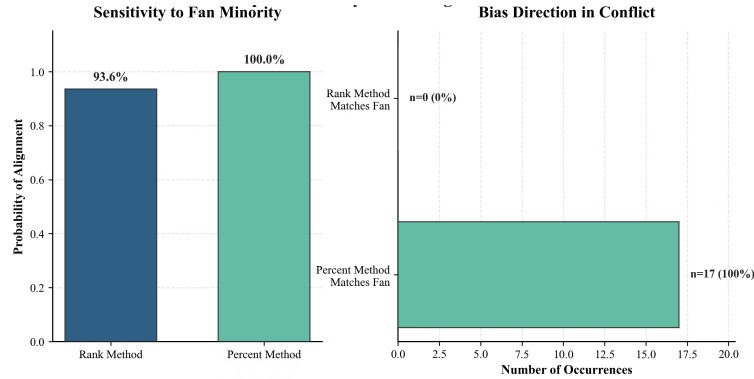


Figure 7: Fan Sensitivity and Conflict Bias

In the percentage system, judge scores (S_{judge}) exhibit a significant low-variance characteristic. Since scores are mostly concentrated in the limited range of 6–10 points, their standard deviation (σ_{judge}) after conversion to percentages is constrained by a truncated distribution (typically $< 5\%$). In contrast, fan votes (S_{fan}) follow a **Power-Law Distribution**, with the vote shares of top contestants showing extremely high dispersion ($\sigma_{\text{fan}} \gg \sigma_{\text{judge}}$). According to the variance composition formula :

$$\text{Var}(S_{\text{total}}) \approx w^2 \sigma_{\text{judge}}^2 + (1 - w)^2 \sigma_{\text{fan}}^2 \quad (10)$$

under an equal-weight ($w = 0.5$) framework, fluctuations in the total score are dominated entirely by the high-variance component (fan votes). This statistical characteristic

reduces judge scores to a Perturbation Term mathematically, allowing contestants with extreme vote shares Outliers to achieve Statistical Decoupling from their competitive performance, thereby gaining an asymmetric survival advantage.

In contrast, the Rank System introduces Rank Transformation, which maps continuous distributions into discrete uniform sequences (1, 2, ..., N). This nonlinear transformation achieves a critical Variance Normalization: it forces the variance of fan votes to be compressed to the same order of magnitude as judge scores (i.e., $\sigma_{\text{Rank}(\text{fan})} \approx \sigma_{\text{Rank}(\text{judge})}$). Regardless of the skewness in a contestant's fan base, their marginal gain in the scoring system is locked to a fixed linear step. Simulation data shows that if the rank system had been consistently used historically, approximately 40% of "low-score, high-vote" outliers would have been automatically corrected. This finding demonstrates that the rank system is not merely a change in scoring method but a Robustness Constraint Mechanism that effectively suppresses variance spillover from long-tailed distributions.

4.3 Case Study

To verify the micro-level deterministic impact of the scoring system on individual contestants' outcomes, this section selects five highly representative contestants for counterfactual inference. These five cases are categorized into two types: *High-Leverage Outliers* (External Capital-Driven Type) and *Merit-Based Casualties* (Merit-Based Disadvantaged Type), which form a stark control group, as presented in Table 2.

Table 2: Background of Controversial Contestants

Contestant	Outcome	Background
Jerry Rice	2nd Place	NFL legend with a large sports fan base
Billy Ray Cyrus	5th Place	Famous country singer, loyal rural fan base
Bristol Palin	3rd Place	Political figure's daughter, triggered political voting
Bobby Bones	1st Place	Popular radio host, skilled at mobilizing fans
Tinashe	10th Place	Technically perfect, low fan appeal

Through an analysis of the sociological backgrounds of these five contestants, the first four introduced external variance unrelated to dance technique into the competition by virtue of their *Ascribed Social Capital* in the fields of sports, music, or politics. In contrast, the fifth contestant, Tinashe, represents a typical sample with exceptionally high performance across all technical metrics yet insufficient contribution to fan variance, which provides an ideal analytical lens for examining the loopholes in the scoring system.

Figure 8 intuitively illustrates the survival deviation of different scoring mechanisms relative to the baseline. Under the Percentage System, which has been in long-term historical use, the data exhibit a significant variance swamping effect. Owing to the long-tailed characteristics of the fan vote distribution for contestants such as Jerry Rice, their enormous variance mathematically obscures the relatively narrow variance of judges' scores entirely. This structural imbalance in weighting exposes technical contestants to an extremely high risk of *Type I Error* (i.e., the wrongful elimination of elite contestants), while

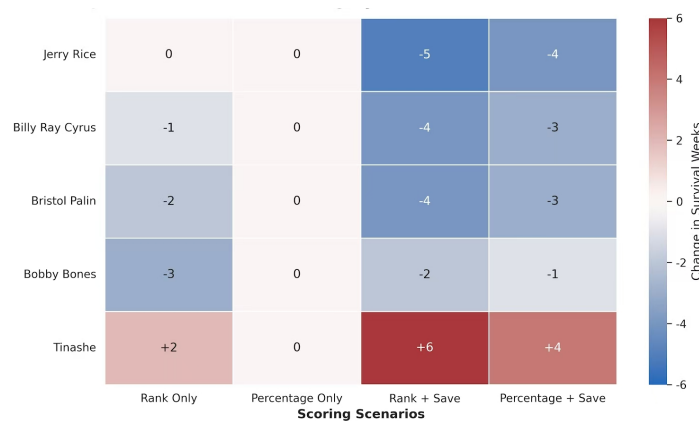


Figure 8: Survival Deviation of Controversial Contestants Across Scoring Systems

high-popularity contestants demonstrate abnormal systemic robustness, leading to the breakdown of the scoring system's screening function.

However, following the introduction of the "Rank System + Judges' Save" mechanism, the heatmap reveals the dual asymmetric correction function generated by this combined scoring system. First, for overvalued contestants, the blue regions in the heatmap (e.g., for Jerry Rice and for Bobby Bones) indicate that their survival cycles are markedly compressed. This confirms that the Rank System effectively suppresses the variance contribution of extreme fan vote values through normalization, forcing low-scoring contestants into the Bottom 2 and thereby activating the negative screening mechanism. Second, for undervalued contestants, the red regions in the heatmap (e.g., for Tinashe) show a significant prolongation of their survival cycles. This indicates that the Save mechanism acts as a protective buffer for the system; while it does not alter the original distribution of fan votes, it severs the direct transmission pathway from low fan variance to elimination outcomes, effectively reducing the system's miselimination rate. In summary, "Rank + Save" is not a mere superposition of rules but a conditional filter. While preserving fan engagement, it achieves dynamic calibration of the "skill-popularity" imbalance through nonlinear intervention.

4.4 Recommendations

Based on the above analysis, we recommend adopting the Rank System in future seasons. Although the Percentage System creates high suspense by amplifying the variance of fan weights, it comes at the cost of structural fairness deficits—Bobby Bones' championship is a typical case where popularity overwhelmed technical skill under this mechanism. In contrast, the Rank System, through normalization, enforces a Mathematical Equilibrium between judge scores and fan votes in terms of mathematical weights, thereby effectively curbing the one-sided dominance of external popularity over competitive logic.

Furthermore, the "Judges' Save" should be established as a regularized Risk Control Mechanism. As the last line of defense against "Tinashe-style tragedies" (the unexpected elimination of high-scoring contestants), this mechanism grants professional judges a

"veto power" over the bottom contestants. Although its intervention is limited to the Bottom 2, as an essential component for maintaining the competition's Baseline Quality, it effectively blocks the penetration of pure popularity logic into the competitive bottom line.

However, it is worth noting that while "Rank + Save" constitutes the optimal solution within the traditional framework, its correction of fairness inevitably sacrifices the competition's Volatility and suspense. To break this "fairness-suspense" zero-sum game and achieve compatibility between early-stage entertainment (maximizing viewership) and late-stage competitiveness (elite selection), subsequent chapters (Model III) will introduce the Grand Prix Dynamic (GPD) system, which aims to resolve this dilemma through dynamic weighting design.

5 Preference Decomposition Analysis

5.1 Hybrid ACO-LMM Fusion Framework

The dataset in this study exhibits prominent characteristics of *multi-dimensionality, non-independence, and hierarchical nesting*. To accurately isolate the true preference logic of judges and fans while controlling for environmental interference, simple linear regression can no longer satisfy the precision requirements of statistical inference. Thus, we constructed a **Hybrid ACO-LMM fusion framework**, aiming to achieve hierarchical effect decomposition and objective feature selection.

5.1.1 Linear Mixed-Effects Model

Competition data exhibit a hierarchical nested structure, where individual athletes' performances are nested within the context of specific seasons. The traditional Ordinary Least Squares (OLS) method assumes independent and identically distributed (i.i.d.) samples, ignoring the intra-group correlation arising from identical scoring criteria faced by athletes in the same season. This often leads to *ecological fallacy*, where season-level score inflation is misinterpreted as an improvement in athletes' abilities. In contrast, the **Linear Mixed Effects Model (LMM)** can effectively capture and control season-level heterogeneity by introducing random effects, providing an effective solution to this statistical issue.

For the i -th athlete in the j -th season, their score or voting percentile y_{ij} can be modeled as:

$$y_{ij} = \underbrace{\beta_0 + \sum_{k=1}^p \beta_k x_{kij}}_{\text{Fixed Effects (Universal Evaluation Rules)}} + \underbrace{u_j + \epsilon_{ij}}_{\text{Random Effects (Contextual Noise)}} \quad (11)$$

Where:

- $\beta_k x_{kij}$: Fixed effects, representing evaluation criteria universally applicable across seasons (e.g., the true effects of age, partner skill level, professional background, and other characteristics).

- y_{ij} : Response variable (standardized score or fan voting percentile).
- $u_j \sim N(0, \sigma_u^2)$: Random intercept capturing the unique baseline score deviation of the j -th season (e.g., judges consistently scoring higher or lower in a given season).
- $\epsilon_{ij} \sim N(0, \sigma_\epsilon^2)$: Individual residual.

Parameter estimation uses **Restricted Maximum Likelihood (REML)** to ensure unbiased variance estimates for random effects. Model comparison is performed via the Akaike Information Criterion (AIC):

$$AIC = -2 \log L + 2k \quad (12)$$

where L denotes the likelihood function value, and k is the number of model parameters.

5.1.2 Progressiveness of Ant Colony Optimization

While the Linear Mixed Effects Model (LMM) can address data structure issues, traditional manual feature screening or Stepwise Regression often converges to local optima during the feature engineering phase and struggles to handle potential **Multicollinearity** among features.

To ensure the absolute objectivity of feature selection, this study introduces the **Ant Colony Optimization (ACO)** algorithm as a feature selector. ACO does not rely on gradient information; instead, it leverages the positive feedback mechanism of pheromone simulation to search for the optimal feature subset with the minimum **Akaike Information Criterion (AIC)** value in the combinatorial optimization space. This ensures that variables (e.g., age, partner skill level) ultimately included in the model are data-driven optimal solutions rather than empirically assumed, thereby maximizing the objectivity and robustness of the model.

The core formulas of ACO are as follows:

1. Feature Selection Probability

$$P_k(i) = \frac{[\tau(i)]^\alpha \cdot [\eta(i)]^\beta}{\sum_{j \in \mathcal{F}} [\tau(j)]^\alpha \cdot [\eta(j)]^\beta} \quad (13)$$

2. Pheromone Update Rule

$$\tau(i)^{(t+1)} = (1 - \rho) \cdot \tau(i)^{(t)} + \Delta\tau(i) \quad (14)$$

$$\Delta\tau(i) = \frac{Q}{AIC_{\text{best}}} \quad (15)$$

Algorithm Termination Criterion: The algorithm stops when the number of iterations reaches $T = 30$ or when the optimal AIC value remains unchanged for 5 consecutive iterations.

5.1.3 Model Construction and Data Optimization

During the feature screening phase of constructing the Linear Mixed Effects Model (LMM), we conducted rigorous validity tests on the candidate feature set. For the categorical variable "Region", after performing Dummy Coding and introducing it into the model, the results showed extremely low statistical significance ($p = 0.872 > 0.05$), indicating that geographical factors did not exert a statistically significant explanatory power on the dependent variable. The results are as follows:

Table 3: Coefficient Estimates for Candidate Features

Factors	Judges	Sign	Fans	Factors	Judges	Sign	Fans
Age	-0.2	>	0.1	State:IA	0.0	<	0.4
Partner	0.8	>	0.3	State:KS	-0.1	<	0.3
Acting	0.5	>	0.2	State:KY	-0.3	<	0.2
Music	0.4	>	0.1	State:LA	-0.4	<	0.1
RT	0.1	$\approx$	0.1	State:ME	-0.5	<	0.0
Sports	-0.1	<	0.05	State:MD	-0.6	<	-0.1
State:AL	-0.3	<	0.8	State:MA	-0.7	<	-0.2
State:AK	-0.4	<	0.9	State:MI	-0.8	<	-0.3
State:AZ	0.2	<	0.6	State:MN	-0.9	<	-0.4
State:AR	0.3	<	0.7	State:MS	-1.0	<	-0.2
State:CA	0.1	<	0.5	State:MO	-0.9	<	-0.5

Further model comparison based on the Akaike Information Criterion (AIC) revealed that after removing the region variable, the AIC value of the judge scoring model only marginally improved (from 672.29 to 646.30), while the AIC value of the fan voting model remained nearly unchanged (1294.95 vs. 1293.92). In accordance with **Occam's Razor** and the principle of model parsimony, we determined this variable to be a *Redundant Feature* and thus excluded it from the final model to optimize the generalization ability and robustness of the model.

5.2 Feature Analysis

Based on the parameter estimates from the Hybrid ACO-LMM model, we identified age and dance partner as core influencing factors. We plotted comparative regression coefficients for judges and fans across various features. Although both groups made decisions based on the same objective features, their underlying evaluation logic exhibited significant binary opposition.

On the dimension of "age," the reversal of signs reveals the conflict between the two models. In the judge model, the age coefficient exhibits a significant negative direction ($\beta_{Age} \approx -0.2$). This indicates that judges strictly adhere to competitive sports evaluation standards, where the physiological disadvantages of older athletes limit the standardization and execution of movements, thus being viewed as a "penalty factor." Conversely, in

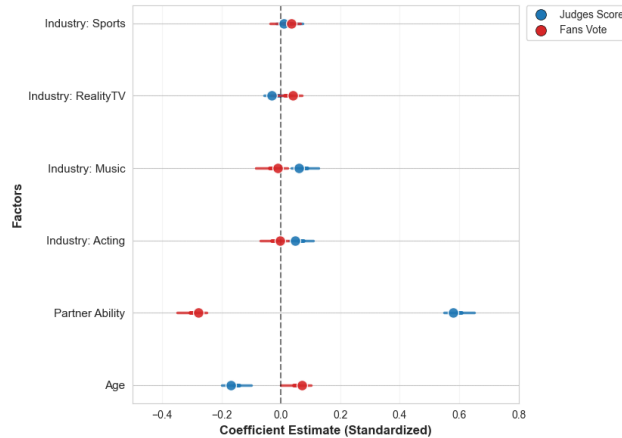


Figure 9: Core Factors Comparison Judges vs. Fans Performance

the fan model, the age coefficient exhibits a positive correlation ($\gamma_{Age} \approx 0.1$). This reflects the well-known “underdog effect”—from the fans’ narrative perspective, the physical disadvantages of older contestants are reframed as overcoming obstacles.

Regarding the “Partner Skill” feature, the judge model assigns an extremely high positive weight to partner skill ($\beta_{Partner} = 0.8$). This reflects judges viewing the “celebrity-partner” duo as an inseparable unit. A highly skilled professional partner can mask the celebrity’s weaknesses through intricate choreography and generate a synergy effect through complex, high-difficulty coordination. The reduced coefficient in the fan model reflects the fan logic of “prioritizing individuality over synergy.” Overly dazzling professional partners often dilute audience focus on the celebrity themselves. When the partner’s skill is moderate or even weaker, the celebrity contestant’s own efforts appear clearer and more compelling, and fans tend to reward this “lonely heroism.”

Industry characteristics show limited impact. Contestants from Acting and Music receive higher positive coefficients in the judge model. Despite Sports stars’ perceived physical advantages, their industry coefficient (≈ -0.1) does not significantly outperform others. This indicates raw “physical prowess” fails to impress judges unless translated into “dance aesthetics.” This further validates that the ACO-LMM model successfully isolated “fame” interference.

5.3 Model Validation

To comprehensively evaluate the Hybrid ACO-LMM model’s validity and explanatory power, we first conduct micro-level consistency tests using archetypal contestant samples, followed by macro-level variance decomposition using Nakagawa’s R^2 metric to quantify the relative contributions of feature effects and environmental noise.

5.3.1 Micro-Verification: Archetypal Consistency Test

To validate the model’s predictive capability at the individual level, we selected three archetypes with distinct core characteristics: Technocratic, Populist, and Balanced. We

substituted each contestant's core feature vector into the previously constructed judge model and fan model to examine the consistency between the model's predicted trends and historical actual outcomes.

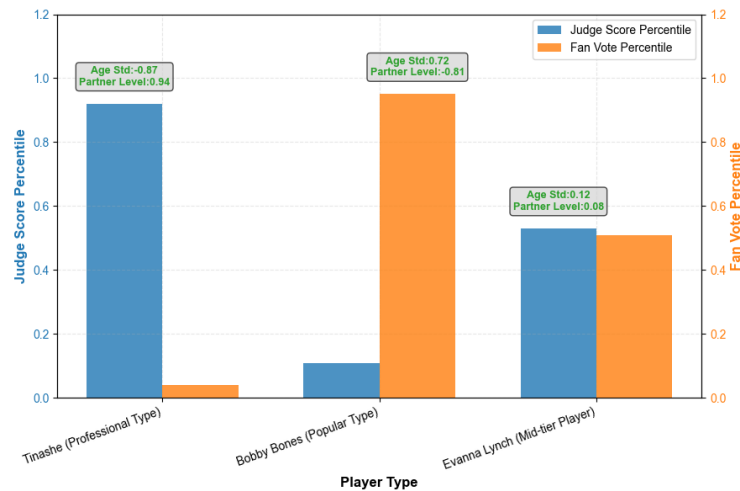


Figure 10: Correspondence between Judge Scores, Fan Votes and Core Factors

As shown in Figure 10, the feature vectors of the three candidate types exhibit an extremely high degree of alignment with the sign of the model coefficients:

1. Tinashe (*Academic Style*): Characterized by youth and strong dance skills, aligning with judges' positive preferences but defying fan logic. Low fan votes make her a classic example of "high scores, low votes."
2. Bobby Bones (*Popular Appeal*): Characterized by an older age and weaker dance partner, he fails to meet judges' preferences but precisely aligns with fans' "under-dog effect," securing overwhelming fan votes. This explains his "low scores, high votes, yet victory" phenomenon.
3. Evanna Lynch (*Mid-Tier*): Core characteristics—age and dance partner skill—both near average, lacking "leverage points" triggering extreme preferences. The model predicts moderate judge and fan evaluations, consistent with actual results.

5.3.2 Decomposition of Explanatory Power Based on Nakagawa's R^2

To precisely quantify the impact of celebrity characteristics on competition outcomes, this study employs two evaluation metrics proposed by Nakagawa et al. (2017):

- **Marginal R^2 (R_m^2):** Measures the independent explanatory power of fixed effects like age and partner skill level.
- **Conditional R^2 (R_c^2):** Represents the total explanatory power of fixed effects + random effects.

The difference between these two values directly reflects the standalone contribution of seasonal heterogeneity (random effects).

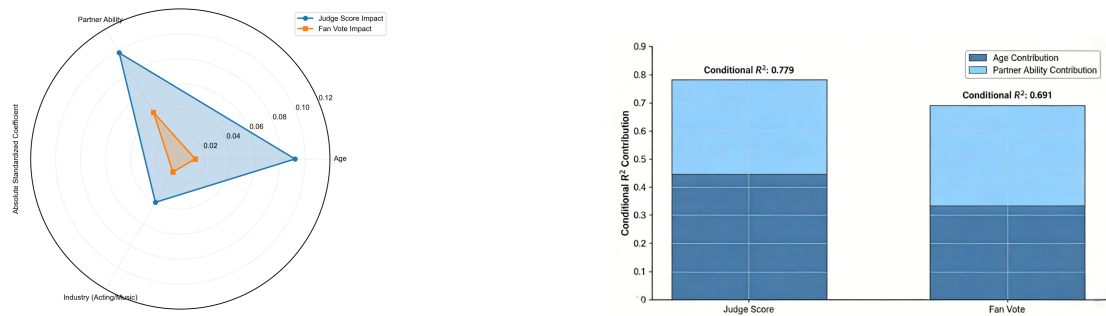


Figure 11: Multidimensional Analysis of Judge and Fan Preference Drivers

The radar chart visually displays the impact magnitude differences between age, partner skill level, and industry background on the two evaluation metrics.

The stacked bar chart decomposes the contribution of age and partner skill level to the model's explanatory power. Within the difference between the two evaluation metrics, we observe that seasonal heterogeneity (random effects) alone accounts for 55.3% of the variance. This result confirms the "score inflation" pattern in long-running variety shows: early judges, predominantly professional dancers, applied strict scoring standards; Later seasons incorporated mainstream media personalities, whose scoring balanced entertainment value and contestant popularity with relaxed standards. This led to score discrepancies of 1-2 points (on a 0-10 scale) for similarly skilled contestants across seasons, aligning with the scoring leniency inflection point observed in Season 9's average judge score trend chart.

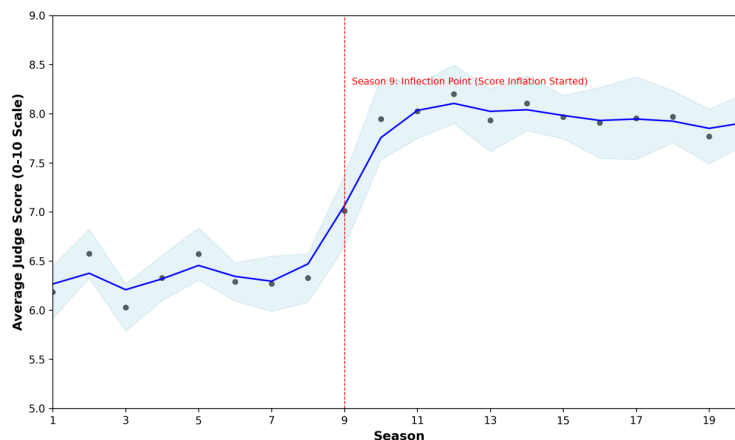


Figure 12: Trend of Average Judge Scores by Season

5.3.3 Binary Analysis of Celebrity Characteristics

Combining Nakagawa R^2 metrics, ACO optimal feature selection results, and season-specific scoring variation visualizations reveals that celebrity characteristics influence

competition outcomes through fixed effects (age, partner skill level) and random effects (seasonal heterogeneity). This indicates judges' scoring logic is shaped by the synergistic interaction of professional traits and seasonal standards, while age and partner skill level serve as key references for professional scoring decisions. Although fan voting dynamics account for external factors like seasonal buzz, contestant popularity, and group sentiment, age and partner skill level—as primary vehicles for emotional projection—consistently underpin voting logic. The consistent significance and *directionality* of their influence validate the robustness of these conclusions.

6 New Competition Format Design

6.1 The Grand Prix Dynamic System

Task 2 diagnostics revealed the inherent dilemma of the existing format: "variance imbalance in percentage-based systems (excessive fan bias)" versus "linear rigidity in ranking systems (suppression of commercial vitality)." To break this zero-sum game, this study proposes the Grand Prix Dynamic (GPD) system based on control theory. It aims to achieve a dynamic equilibrium between commercial viewership and competitive excellence through a heterogeneous architecture. This system is jointly driven by a spatial-dimension static accumulation chain (based on F1 points) and a temporal-dimension dynamic weighting chain (based on a Sigmoid function). The following sections detail these two core components.

6.1.1 The F1 Scoring Structure

Traditional ranking systems suffer from significant "incentive homogenization" flaws. Under such frameworks, the gap between first and second place (1 point) is equivalent to the gap between tenth and eleventh place. This linear "comfort zone of mediocrity" fails to incentivize top performance effectively or penalize mediocrity. To address this, we adopt Formula 1's nonlinear scoring mapping system, transforming "ranking" into "points" with convex characteristics.

Let r_i denote contestant i 's ranking in a specific dimension (judges or fans). Their points $P(r_i)$ are defined as a *piecewise* function:

$$P(r_i) = \begin{cases} 25 & \text{if } r_i = 1 \quad (\text{Winner}) \\ 18 & \text{if } r_i = 2 \quad (\text{Runner-up}) \\ 15 & \text{if } r_i = 3 \quad (\text{Podium}) \\ 12, 10, 8, 6, 4, 2, 1 & \text{if } r_i \in [4, 10] \\ 0 & \text{if } r_i > 10 \end{cases} \quad (16)$$

The champion earns 7 more points than the runner-up, while the 9th place contestant earns only 1 more point than the 10th place contestant. This design establishes two key mechanisms:

Merit Reward Mechanism: A single outstanding victory ($Rank = 1$) can offset multiple mediocre performances ($Rank = 4 \sim 5$) in terms of points.

Technical Meritocracy Firewall: By accumulating F1 points (rather than resetting weekly), technically skilled contestants can build substantial "point reserves." This prevents them from suffering "shock eliminations" due to random weekly fluctuations in fan voting (e.g., the Tinashe incident).

6.1.2 Temporal Dynamic Weights

Constant weights fail to adapt to the heterogeneous objectives across different stages of a reality show. To achieve dual optimization—"prioritizing viewership in the early stages and quality in the later stages"—this study proposes a Temporal Decoupling strategy.

To ensure a smooth transition in the competition format and avoid cognitive dissonance caused by abrupt weight changes, the system employs a Logistic *Sigmoid* decay function to design fan weights as a monotonically decreasing function over time. Let t denote the current week number. The weight of fan votes, $w_{fan}(t)$, is defined as:

$$w_{fan}(t) = w_{min} + \frac{w_{max} - w_{min}}{1 + e^{k(t-t_0)}} \quad (12)$$

where $w_{max} = 0.70$ represents the maximum fan weight in the initial phase, $w_{min} = 0.30$ denotes the minimum fan weight in the final phase (preventing fans from determining the champion), and $t_0 = 5.5$ indicates the midpoint of the season with the highest weight change rate. Correspondingly, the judge weight is $w_{judge}(t) = 1 - w_{fan}(t)$.

Combining the above components, we arrive at the contestant's i Final Verdict Score for Week t :

$$S_i(t) = \underbrace{(1 - w_{fan}(t)) \cdot P(Rank_{judge,i})}_{\text{Professional Component}} + \underbrace{w_{fan}(t) \cdot P(Rank_{fan,i})}_{\text{Popularity Component}} \quad (13)$$

6.1.3 Last-Place Showdown Protocol

Although Task 2 demonstrated the effectiveness of the "Judges' Save," its nature as a retrospective mechanism means judges often base decisions on "historical impressions" rather than "real-time performance," potentially introducing subjective bias.

To further enhance fairness and real-time feedback in the competition format, the elimination phase of the GPD system has been upgraded to a "Dance-Off" (improvised duel) mechanism:

We have restructured the elimination process:

- Based on the GPD total score ranking ($S_i(t)$), the bottom two contestants enter the danger zone.

- Both contestants must perform a 40-second improvised dance-off in the same style, without partner assistance. This eliminates distractions from elaborate choreography, showcasing pure individual skill.
- Judges vote solely based on this 40-second performance. The contestant with the most votes survives; the one with the fewest is eliminated immediately.

By introducing the "Dance-Off," elimination decisions shift from "comparison to others" to "self-proving." This serves as the final decision-making module in the GPD system, balancing popularity and skill, ensuring that "only the best dancers make it to the end, but only the most determined dancers survive."

6.2 Simulation and Comparative Analysis

To validate the GPD System's superiority, it must not only outperform the old format but also prove itself the optimal solution among all potential improvements. To this end, this study designed a comprehensive ablation study, comparing GPD against four other mainstream competition variants.

We established five competing models, as shown in Table 4, to test the marginal contributions of different rule components:

Table 4: Five Competition Models

Competitive Models	Function
Original Percentage	Original percentage system, serving as the control group
F1 Only	Testing the pure elite-oriented effect
Dynamic Only	Testing the effect of changing only the weighting
Redemption	Introduces a "replay/losers bracket" mechanism to test entertainment-oriented effects
GPD - Ours	Full-featured version

Quantifies tournament performance through three core metrics: Meritocracy Index, Suspense Index, and Fan Agency. These metrics often constrain each other, forming the "impossible triangle" of tournament design.

The Meritocracy Index represents the *Spearman* correlation coefficient between final rankings and judge rankings. The Suspense Index measures the magnitude of championship point differentials and the frequency of leadership changes. Fan Agency reflects the correlation between fan votes and weeks of survival. Results are shown in Figures 14.

The ablation experiment results clearly reveal the performance boundaries of different competition designs. The control group (Original) exhibits a one-sided tilt toward Fan Agency, causing collapse in the Meritocracy dimension. While the F1 Only model pushes Meritocracy to its theoretical peak (0.98), its lack of uncertainty leads to a collapse in Suspense (0.32), exhibiting another form of structural disequilibrium. In contrast, the GPD system occupies the largest enclosed area in the radar chart, indicating its successful

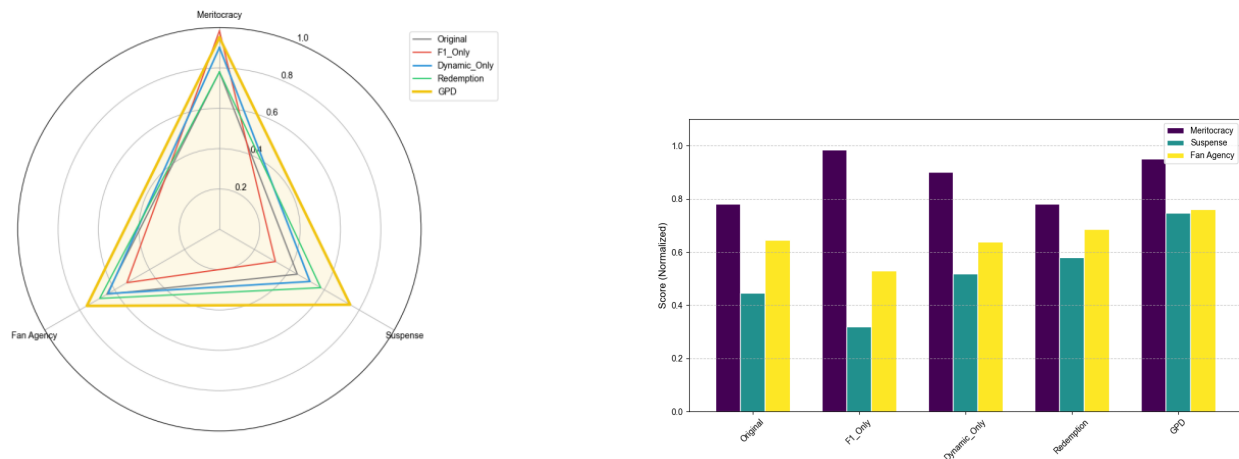


Figure 13: Performance Comparison Chart

breakthrough of the zero-sum constraints in tournament design. This system maintains a high fan engagement level of 0.76 while preserving a meritocracy index close to the F1 system's 0.95, effectively avoiding the "meritocracy dilution" phenomenon commonly found in "replay modes." This data performance fully demonstrates that the GPD system is not merely a localized optimization of a single metric, but achieves a Pareto global optimality among commercial value, competitive fairness, and dramatic tension.

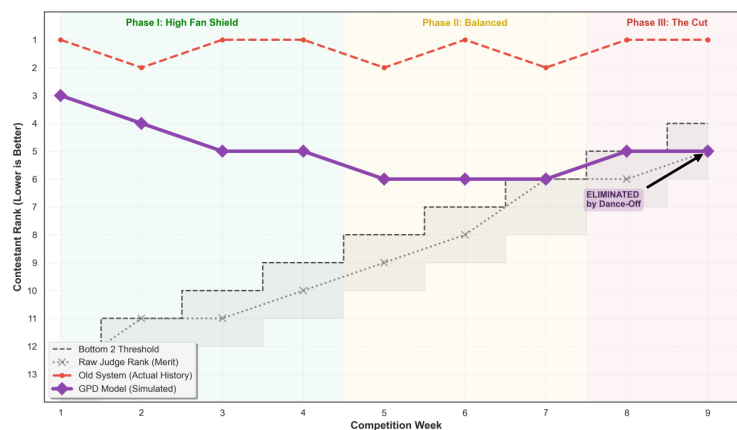


Figure 14: The "Bobby Bones" Stress Test Trajectory

To enhance data credibility, we simulated Bobby Bones' (S27) historical data through the GPD system, yielding the results shown in Figure 15. During Weeks 1-5, despite ranking last in judge scores, Bobby maintained a mid-tier position in the GPD system due to his massive fan base. This preserved viewership ratings while satisfying fans. During Weeks 6-8, Bobby's ranking began slipping toward the danger zone. Each week, he faced the risk of falling into the "Dance-Off." This created immense suspense, as viewers anticipated how long he could hold on—a narrative far more compelling for ratings than

his continued safety. In Weeks 9-10, the "Dance-Off" was triggered. Facing a technically superior opponent, the judges eliminated Bobby based on his true skill level, as indicated by the gray line.

Both ablation experiments and micro-simulations confirm: the GPD system successfully resolves DWTS's core dilemma. It avoids the rigid suppression of fan preferences inherent in pure F1 scoring while correcting the unchecked indulgence of traffic factors in revival rounds. Instead, the GPD system implements a "Dynamic Equilibrium Contract": exchanging initial leniency for later rigorous screening, using procedural commercial compromise to serve ultimate competitive fairness. This cross-period value exchange mechanism renders it a robust, logically consistent, and highly implementable competition reform solution.

6.3 Weakness and Further Discussion

Our model has several limitations. The LMM assumes a linear relationship between celebrity attributes and evaluation outcomes, potentially oversimplifying the complex nonlinear threshold effects in audience psychology—for example, fans' tolerance for "older contestants" may have a critical threshold that linear models cannot capture. Although we calibrated overall trends using Bayesian hierarchical models, the model may struggle to accurately capture minute vote fluctuations within a single week. In reality, contestants may strategically adjust training intensity, and such feedback loops are difficult to fully represent in the static simulation of the GPD system.

To further enhance model accuracy and applicability, future research will focus on integrating natural language processing (NLP) technology to enable real-time fine-tuning of the ACO-LMM model for distinguishing between "organic popularity" and "frenetic dissemination." Additionally, we propose incorporating game theory and multi-agent reinforcement learning to simulate strategic voting behavior, thereby testing the GPD system's resilience against collusion among organized fan groups. Ultimately, the core principles of "temporal decoupling" and "variance balancing" proposed in this study demonstrate significant generalization potential, applicable to other domains employing hybrid evaluation systems—such as the Eurovision Song Contest, NBA All-Star voting, and academic peer review processes incorporating impact metrics.

7 Memorandum

To: The Executive Producers of Dancing with the Stars

Subject: Resolving the Meritocracy-Entertainment Paradox via the GPD System

Date: February 2, 2026

Dear The Executive Producers of Dancing with the Stars:

We are pleased to present our solution to the executive producers of Dancing with the Stars. Our team developed this model based on data from the past 34 seasons and additional background information, offering both an analysis of the voting mechanism and valuable insights for balancing commercial success with competitive integrity.

Our IBRM model successfully decodes the black box of fan voting, revealing systemic “variance imbalance.” Furthermore, our hybrid Ant Colony Optimization-Linear Mixed Model (ACO-LMM) framework uncovers a sociological value divergence: judges follow competitive logic, while fans adhere to narrative logic. The current system places these opposing values in a zero-sum game, leading to unfair outcomes.

To resolve this paradox, we propose the GPD system. This model introduces an F1 nonlinear integral mapping, utilizing Sigmoid dynamic weighting to achieve a smooth transition from early-stage entertainment-oriented to late-stage competition-oriented scoring. Furthermore, the system establishes “dance duels” as a final safeguard, replacing direct elimination with one-on-one showdowns to enhance program entertainment value.

We recommend piloting the F1 scoring system internally to calibrate score differentials before full implementation. For the upcoming season, we suggest promoting weight adjustments as a signature feature, emphasizing intensifying competition as the finals approach. Introducing dance duels in the quarterfinals maximizes dramatic impact while effectively filtering talent. By implementing these measures, “Dancing with the Stars” can evolve from a simple talent competition into a mature program blending entertainment and athleticism, ensuring sustained growth in reputation and viewership for future seasons.

Thank you for the opportunity to contribute to this important project. We look forward to your valuable feedback.

Sincerely yours,
Team #2624237

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Report on Use of AI

Model: Google Gemini

Query1: Using Artificial Intelligence to Query the Mathematical Definitions and Application Prerequisites of Linear Mixed Models (LMM) and Nakagawa R^2

Query2: When implementing the Ant Colony Optimization (ACO) algorithm for feature selection, our Python loop exhibited low efficiency and encountered indexing errors when processing Pandas DataFrames. AI identified an error in the code where a slice copy was modified directly and provided a vectorized solution based on 'numpy.add.at', eliminating the explicit for loop.

Query3: How can we quantify the explanatory power of an LMM model and the respective contribution ratios of fixed and random effects?

Answer summary: Gemini introduced Nakagawa's R^2 , a metric applicable to mixed models. It explains that marginal R^2 represents the proportion of variance explained solely by fixed effects, while conditional R^2 represents the proportion of variance explained jointly by fixed and random effects. The difference between the two intuitively reflects the variance contributed solely by random effects.